

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281346

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Suarez; Carlos et al.

HEALTH MANAGEMENT SYSTEM

Abstract

A therapeutic wrap includes a flexible member configured to wrap around an affected area. Actuators are operably coupled to the flexible member. Each actuator is operable between a deployed state and a non-deployed state. A first coupling feature is disposed proximate a first edge on a first surface of the flexible member. A second coupling feature is disposed proximate a second edge on a second surface of the flexible member. The first coupling feature is configured to engage the second coupling feature when the flexible member is wrapped around the affected area and when the flexible member is slidably adjusted relative to the affected area. A retaining feature is coupled to at least one end of the flexible member and configured to retain the flexible member in a selected position on the affected area.

Inventors: Suarez; Carlos (Syracuse, NY), Kellner; David (Auburn, NY), Lane; John (Weesport, NY), Burnham; Aaron R. (Auburn, NY), Bremer; Edward (Penfield, NY), Nigolian; Tyler (Syracuse, NY), Lia; Raymond (Auburn, NY), Vivenzio; Robert L. (Auburn, NY), Wawro; Thaddeus J. (Auburn, NY), Douglas; Rachel K. (Latrobe, PA)

Applicant: Welch Allyn, Inc. (Skaneateles Falls, NY)

Family ID: 80224639

Appl. No.: 19/216891

Filed: May 23, 2025

Related U.S. Application Data

parent US division 17341742 20210608 parent-grant-document US 12343305 child US 19216891
us-provisional-application US 63064486 20200812

Publication Classification

Int. Cl.: A61H9/00 (20060101)

U.S. Cl.:

CPC **A61H9/0078** (20130101); **A61H9/0071** (20130101); A61H2201/0103 (20130101);
A61H2201/1215 (20130101); A61H2201/1635 (20130101); A61H2201/164 (20130101);
A61H2201/165 (20130101); A61H2201/5097 (20130101); A61H2205/06 (20130101);
A61H2205/106 (20130101); A61H2209/00 (20130101); A61H2230/04 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a division of U.S. patent application Ser. No. 17/341,742, filed on Jun. 8, 2021, entitled “HEALTH MANAGEMENT SYSTEM,” which claims priority to and the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Application No. 63/064,486, filed on Aug. 12, 2020, entitled “HEALTH MANAGEMENT SYSTEM,” the disclosure of which is hereby incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to a health management system, and more particularly, a health management system for treating various conditions, including lymphedema and deep vein thrombosis.

SUMMARY OF THE DISCLOSURE

[0003] According to one aspect of the present disclosure, a wrap for covering an affected area includes a flexible member configured to wrap around the affected area. Actuators are operably coupled to the flexible member. Each actuator is operable between a deployed state and a non-deployed state. A first coupling feature is disposed proximate a first edge on a first surface of the flexible member. A second coupling feature is disposed proximate a second edge on a second surface of the flexible member. The first coupling feature is configured to engage the second coupling feature when the flexible member is wrapped around the affected area and when the flexible member is slidably adjusted relative to the affected area. A retaining feature is coupled to at least one end of the flexible member and is configured to retain the flexible member in a selected position on the affected area.

[0004] According to another aspect of the present disclosure, a garment includes a flexible member configured to be worn on an affected area. A power source is operably coupled to the flexible member. An electromagnetic actuator is operably coupled to the power source. The electromagnetic actuator is operable between a deployed state and a non-deployed state in response to a voltage from the power source. The electromagnetic actuator applies pressure to the affected area when in the deployed state.

[0005] According to another aspect of the present disclosure, a health management system includes a therapeutic pad for supporting an affected area of a patient. Actuators are operably coupled to the therapeutic pad. A controller is communicatively coupled to the actuators. The controller is configured to adjust the actuators between a deployed state and a non-deployed state. The actuators are configured to apply pressure to the affected area supported by the therapeutic pad when the actuators are in the deployed state.

[0006] These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] In the drawings:

[0008] FIG. 1 is a side perspective view of a therapeutic wrap worn on an arm of a patient, according to the present disclosure;

[0009] FIG. 2 is a partial top perspective view of the therapeutic wrap of FIG. 1 partially applied to the arm of the patient, according to the present disclosure;

[0010] FIG. 3 is a schematic view of a therapeutic wrap worn on an arm of a patient, according to the present disclosure;

[0011] FIG. 4 is a top plan view of a therapeutic wrap, according to the present disclosure;

[0012] FIG. 5 is a bottom plan view of the therapeutic wrap of FIG. 4, according to the present disclosure;

[0013] FIG. 6 is a side perspective view of a sock worn on a leg of a patient, according to the present disclosure;

[0014] FIG. 7 is a schematic view of a voice coil actuator in a deployed state affecting pressure on an engagement layer, according to the present disclosure;

[0015] FIG. 8 is a schematic view of a voice coil actuator with a guide member, according to the present disclosure;

[0016] FIG. 9A is a schematic view of a voice coil actuator with a guide member in a deployed state and providing pressure on the skin of a patient, according to the present disclosure;

[0017] FIG. 9B is a schematic view of a voice coil actuator with a guide member in a non-deployed state and spaced from the skin of a patient, according to the present disclosure;

[0018] FIG. 10 is a front view of a slide assembly of a treatment assembly with a portion of a slide illustrated in phantom, according to the present disclosure;

[0019] FIG. 11A is a schematic view of a voice coil actuator in a deployed state operably coupled with a slide assembly in an extended condition with a portion of a slide illustrated in phantom, according to the present disclosure;

[0020] FIG. 11B is a schematic view of a voice coil actuator in a non-deployed state operably coupled with a slide assembly in a retracted condition with a portion of a slide illustrated in phantom, according to the present disclosure;

[0021] FIG. 12 is a side perspective view of a slide assembly of a treatment assembly with a portion of a slide illustrated in phantom, according to the present disclosure;

[0022] FIG. 13 is a schematic view of a pad providing treatment to a patient, according to the present disclosure;

[0023] FIG. 14 is a side perspective view of a pad with massage devices, according to the present disclosure;

[0024] FIG. 15 is a side perspective view of a massage device, according to the present disclosure;

[0025] FIG. 16 is a side perspective view of a massage device, according to the present disclosure;

[0026] FIG. 17 is a side perspective view of a pad that includes bladders, according to the present disclosure;

[0027] FIG. 18 is a schematic side perspective view of a treatment assembly with a sling that includes actuators with the treatment assembly in an open position, according to the present disclosure;

[0028] FIG. 19 is a schematic front elevational view of the treatment assembly of FIG. 18, according to the present disclosure;

[0029] FIG. 20 is a schematic front elevational view of the treatment assembly of FIG. 18 in a closed position and the sling in a relaxed condition, according to the present disclosure;

[0030] FIG. 21 is a schematic front elevational view of the treatment assembly of FIG. 20 with the

sling in a fitted condition, according to the present disclosure;

[0031] FIG. **22** is a schematic side perspective view of a treatment assembly that includes actuators, according to the present disclosure;

[0032] FIG. **23** is a side perspective view of a treatment assembly that includes a garment with bladders and an inflatable therapeutic pad in a deflated state, according to the present disclosure;

[0033] FIG. **24** is a side perspective view of the treatment assembly of FIG. **23** with the therapeutic pad in an inflated state, according to the present disclosure; and

[0034] FIG. **25** is a block diagram of a health management system, according to the present disclosure.

DETAILED DESCRIPTION

[0035] The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a health management system. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

[0036] For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof, shall relate to the disclosure as oriented in FIG. **1**. Unless stated otherwise, the term “front” shall refer to a surface closest to an intended viewer, and the term “rear” shall refer to a surface furthest from the intended viewer. However, it is to be understood that the disclosure may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific structures and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0037] The terms “including,” “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises a . . .” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0038] Referring to FIGS. **1-25**, reference numeral **10** generally designates a health management system **10** that includes a treatment assembly **12** that has a flexible member **14** configured to wrap around an affected area. Actuators **16** are operably coupled to the flexible member **14**. Each actuator **16** is operable between a deployed state and a non-deployed state. A first coupling feature **18** is disposed proximate a first edge **20** on a first surface **22** of the flexible member **14**. A second coupling feature **24** is disposed proximate a second edge **26** on a second surface **28** of the flexible member **14**. The first coupling feature **18** is configured to engage the second coupling feature **24** when the flexible member **14** is wrapped around the affected area and when the flexible member **14** is slidably moved relative to the affected area. A retaining feature **30** is coupled to at least one end of the flexible member **14** and configured to retain the flexible member **14** in a selected position on the affected area.

[0039] The health management system **10** may be used to manage certain health conditions, such as lymphedema, deep vein thrombosis (DVT), skin ulcers, and other similar conditions. Lymphedema is a chronic disease that may result from a variety of factors, including diabetes, radiation, chemotherapy, and surgery. Lymphedema generally causes the body to fill with lymphatic fluid, which results in swelling. The swelling may cause pain and discomfort, as well as cause lesions and hardening of the skin. One method to help manage lymphedema includes massage therapy.

Massage therapy may assist in moving lymphatic fluid to the lymphatic system, and subsequently to the circulatory system. DVT is a condition where blood clots form in parts of the body. DVT often affects people who are sedentary or aged. Massage therapy may assist in increasing blood flow in affected areas to manage DVT. Skin ulcers may be caused by prolonged periods of poor blood flow to the affected area of the body. Poor blood flow may occur as a result of an infection, immobility, or health conditions, such as diabetes. Massage therapy may be utilized to increase local blood flow, which may prevent the development of skin ulcers.

[0040] Massage therapy treatments may be provided by the health management system **10** disclosed herein. The person receiving the treatment is generally referred to herein as a patient. The treatment provided by the health management system **10** may be more convenient for the patient as the mobility of the health management system **10** allows for treatment to be provided in the home or otherwise outside of a traditional medical setting.

[0041] Referring to FIG. **1**, the treatment assembly **12** includes the flexible member **14**, which is configured as a wrap or a therapeutic wrap **38**. The therapeutic wrap **38** is configured to be spiraled or wound around an extent of the affected area, which is illustrated as an arm of the patient. It is contemplated that the therapeutic wrap **38** may be a spiral wrap, as illustrated, or alternatively may be any type of therapeutic device that surrounds or covers the affected area to provide treatment, such as, for example, a sleeve, a sock, another garment, another type of covering, etc. The therapeutic wrap **38** configured as a spiral wrap is elongated to form a helix shape along the arm from proximate the shoulder to proximate the wrist. The flexible member **14** is generally constructed of fabric or similar material to abut the affected area when wound in the helix shape.

[0042] Referring to FIGS. **1** and **2**, the therapeutic wrap **38** includes the first surface **22** and the second surface **28**. When the therapeutic wrap **38** is applied to the affected area, the first surface **22** abuts the skin of the patient (e.g., an inner surface), and the second surface **28** is oriented away from the skin (e.g., an outer surface). With each new spiral, illustrated as spirals **38A-38C**, of the therapeutic wrap **38**, the therapeutic wrap **38** at least partially overlaps a previous spiral to cover the affected area with no gaps between adjacent spirals. Generally, the therapeutic wrap **38** is first applied to the hand or wrist and then spiraled up the arm toward the shoulder.

[0043] To maintain the helix shape of the therapeutic wrap **38**, the therapeutic wrap **38** includes the first coupling feature **18** that is configured to engage the second coupling feature **24**. The first coupling feature **18** extends along the first edge **20** on the first surface **22** of the therapeutic wrap **38**. The second coupling feature **24** extends along the second edge **26** on the second surface **28** of the therapeutic wrap **38**. Accordingly, when the first edge **20** overlaps the second edge **26** during adjacent spirals around the affected area, the first coupling feature **18** is disposed over and engages the second coupling feature **24** of the previous spiral. Each of the first coupling feature **18** and the second coupling feature **24** may be Velcro®, hook-and-loop fasteners, snap features, or any other similar fastening features configured to engage with one another.

[0044] As the therapeutic wrap **38** is wound around the affected area, the second coupling feature **24** is exposed (e.g., facing outward from the affected area) and the first coupling feature **18** is configured to engage the second coupling feature **24** in each subsequent spiral. For example, in the spiral **38A** the second coupling feature **24** is exposed and as the therapeutic wrap **38** winds into the spiral **38B**, the first coupling feature **18** on the spiral **38B** engages the second coupling feature **24** on the spiral **38A**. The first coupling feature **18** and the second coupling feature **24** may engage one another between adjacent spirals (e.g., between spirals **38A**, **38B**, between spirals **38B**, **38C**, etc.) from proximate the wrist to proximate the shoulder. It is generally contemplated that each of the first coupling feature **18** and the second coupling feature **24** extend along the entire longitudinal extent of the therapeutic wrap **38** to provide a customizable fit of the therapeutic wrap **38** around the arm.

[0045] Referring still to FIGS. **1** and **2**, the helix shape of the therapeutic wrap **38** may be maintained by the engagement between the first coupling feature **18** and the second coupling

feature **24** as the patient slidably moves the therapeutic wrap **38** relative to the affected area. The patient may initially wrap the therapeutic wrap **38** around the arm for treatment, as described above. To remove the therapeutic wrap **38**, the patient may pull the therapeutic wrap **38** and slide the therapeutic wrap **38** from the affected area without unwinding the therapeutic wrap **38**. To again apply treatment to the affected area, the patient may insert the arm through the therapeutic wrap **38**, which remains in its original helix shape, and pull the therapeutic wrap **38** onto the affected area without having to rewind the therapeutic wrap **38** around the arm. The patient may slide the preformed therapeutic wrap **38** over the affected area.

[0046] The first coupling feature **18** and the second coupling feature **24** are configured to remain engaged with one another as the therapeutic wrap **38** is slidably moved relative to the affected area (e.g., removed from or applied to the affected area). The therapeutic wrap **38** remaining in the helix shape may be advantageous for a more convenient application of the therapeutic wrap **38** to provide treatment. The patient may not go through the cumbersome process of winding and unwinding the therapeutic wrap **38** for each treatment.

[0047] The therapeutic wrap **38** is adjustable by disengaging the second coupling feature **24** from the first coupling feature **18** and unwinding the therapeutic wrap **38**. Unwinding and rewinding the therapeutic wrap **38** over the affected area may adjust the custom shape and size of the helix shape around the arm. Depending on the patient or the amount of excess fluid in the arm, the therapeutic wrap **38** may be customized and adjusted to form fit around the arm of the patient.

[0048] Referring still to FIGS. **1** and **2**, a distal end **40** of the therapeutic wrap **38** disposed proximate the wrist includes the retaining feature **30**. The retaining feature **30** is configured as an aperture **42** defined in the distal end **40** of the therapeutic wrap **38**, which receives a finger, such as a thumb, of the patient. The patient may slide the thumb through the aperture **42**, wrap the therapeutic wrap **38** over the hand in the spiral **38A** and around the wrist in the spiral **38B**, and then continue to wind the therapeutic wrap **38** around the arm in the spiral **38C** toward the shoulder. Alternatively, the patient may slide the thumb through the aperture **42** and then slide the therapeutic wrap **38** over the affected area. The retaining feature **30** is advantageous for retaining the therapeutic wrap **38** in a selected position on the affected area. For example, the thumb extending through the aperture **42** may prevent the therapeutic wrap **38** from sliding up the arm during treatment.

[0049] The aperture **42** for the thumb may also be advantageous for more conveniently removing the therapeutic wrap **38** from the arm. The patient may grasp the therapeutic wrap **38** adjacent to the aperture **42** and pull to slide the therapeutic wrap **38** from the affected area. The therapeutic wrap **38** may slide from the affected area more easily with the aperture **42** than without the aperture **42**.

[0050] Referring to FIGS. **3-5**, an additional or alternative configuration of the treatment assembly **12** includes the flexible member **14** configured as a wrap or a therapeutic wrap **50**. The therapeutic wrap **50** may be configured as a spiral wrap configured to helically wind around the affected area, such as an arm as illustrated in FIG. **5**, similar to the therapeutic wrap **38** illustrated in FIGS. **1** and **2**. It is contemplated that the therapeutic wrap **50** may be a spiral wrap, as illustrated in FIGS. **3-5**, or alternatively may be any type of therapeutic device that surrounds or covers the affected area to provide treatment, such as, for example, a sleeve, a sock, another garment, another type of covering, etc.

[0051] The therapeutic wrap **50** generally includes the first coupling feature **18** that extends along the first edge **20** of the first surface **22** of the therapeutic wrap **50** and the second coupling feature **24** that extends along the second edge **26** of the second surface **28**. The first and second coupling features **18**, **24** are configured to engage one another to retain the therapeutic wrap **50** in the helix shape when the therapeutic wrap **50** is disposed on the affected area and when the therapeutic wrap **50** is slidably moved relative to the affected area.

[0052] The therapeutic wrap **50** includes two retaining features **30**, which are configured as a first

or distal elongated flap 52 and a second or proximal elongated flap 54. The distal elongated flap 52 extends from the distal end 40 of the therapeutic wrap 50 and the proximal elongated flap 54 extends from a proximal end 56 of the therapeutic wrap 50. The elongated flaps 52, 54 may extend along opposing edges of the therapeutic wrap 50, where the opposing edges extend between the first and second edges 20, 26. The first or distal elongated flap 52 may extend from the first edge 20 and extend beyond the second edge 26. The second or proximal elongated flap 54 may be rotated approximately 90° compared to the first elongated flap 52 and extend from the second edge 26 of the therapeutic wrap 50 and beyond the first edge 20. The elongated flaps 52, 54 are generally arranged parallel to one another and at an oblique angle relative to the longitudinal extent of the therapeutic wrap 50.

[0053] Each elongated flap 52, 54 may include a first fastening member 58 on the first surface 22 and a second fastening member 60 on the second surface 28. The first and second fastening members 58, 60 may be Velcro®, hook-and-loop fasteners, snap features, or any other similar fastening features configured to engage one another. The first and second fastening members 58, 60 of each elongated flap 52, 54 are configured to engage one another to retain the therapeutic wrap 50 in the selected position on the affected area.

[0054] Referring still to FIGS. 3-5, the therapeutic wrap 50 may be placed with the distal elongated flap 52 adjacent to the wrist. The distal elongated flap 52 is configured to circle the wrist and overlap itself. In the illustrated configuration, the second fastening member 60 faces outwardly from the skin of the patient and is engaged by the first fastening member 58 when the distal elongated flap 52 overlaps itself. The alignment of the distal elongated flap 52 on the wrist provides for the angled spiraling of the therapeutic wrap 50. The distal elongated flap 52 operates as a cuff for retaining the distal end 40 of the therapeutic wrap 50 in the selected position on the affected area.

[0055] After the patient has wound the therapeutic wrap 50 around the affected area and reached the second or proximal elongated flap 54, the proximal elongated flap 54 may be utilized to retain the proximal end 56 of the therapeutic wrap 50 in the selected position adjacent to the shoulder of the patient. It is also contemplated that the proximal elongated flap 54 may retain the proximal end 56 of the therapeutic wrap 50 adjacent to the elbow of the patient depending on the configuration of the therapeutic wrap 50. Similar to the distal elongated flap 52, the proximal elongated flap 54 circles the upper arm and overlaps itself to engage the first fastening member 58 with the second fastening member 60. The elongated flaps 52, 54 may be advantageous for securing both the distal end 40 and the proximal end 56 of the therapeutic wrap 50 in the selected position on the arm to apply the treatment to the affected area.

[0056] To remove the therapeutic wrap 50, the patient may disengage the first fastening member 58 from the second fastening member 60 on each of the elongated flaps 52, 54. The patient may then slide the therapeutic wrap 50 from the affected area with the therapeutic wrap 50 maintaining the helix shape via the engagement between the first and second coupling features 18, 24. To reapply the therapeutic wrap 50, the patient can slide the therapeutic wrap 50 over the affected area and engage the elongated flaps 52, 54. To readjust the size of the helix shape, the patient may unwind and rewind the therapeutic wrap 50. While the therapeutic wrap 50 is illustrated on an arm, it is contemplated that the therapeutic wrap 50 may be wrapped around a leg without departing from the teachings herein.

[0057] Referring to FIGS. 1-5, the flexible member 14 of the treatment assembly 12 is generally constructed of two layers of fabric or other material. The two layers are coupled to one another at the first edge 20 and the second edge 26. The actuators 16 in the configurations set forth in FIGS. 1-5 are configured as channels 66A-66C, collectively referred to as channels 66, which extend along the longitudinal extent of the flexible member 14. Generally, the channels 66 are arranged parallel to one another and extend between the distal end 40 and the proximal end 56 of the flexible member 14. However, it is contemplated that the channels 66 may taper approximate the distal end

40 or the proximal end **56** based on the size and shape of the affected area (e.g., the wrist compared to the upper arm). The two layers of the flexible member **14** are coupled to one another between each channel **66** to define the channels **66**.

[0058] Each channel **66** is adjustable between the deployed state and the non-deployed state. The deployed state may be an inflated condition of each channel **66**, and the non-deployed state may be a deflated condition of each channel **66**. When in the deployed state, the channels **66** apply pressure to the affected area.

[0059] It is generally contemplated that each channel **66** is in fluid communication with a pump **68**, which is configured to adjust each channel **66** to the deployed state via a corresponding inlet **70** defined in the proximal end **56** of the flexible member **14**. In the illustrated example, the flexible member **14** defines the three channels **66** that extend the entire longitudinal extent of the flexible member **14** and three inlets **70**. It is contemplated that the channels **66** may be separated into zones along the longitudinal extent of the flexible member **14**. In such configurations, the flexible member **14** may define additional inlets **70** corresponding with each zone.

[0060] Referring still to FIGS. 1-5, the pump **68** may be operably coupled to the proximal end **56** of the flexible member **14** proximate the shoulder of the patient. The pump **68** may direct fluid into one or more of the channels **66**. The fluid may be a gas, a liquid, a gel, or any other fluid. The pump **68** may direct the fluid into the selected channel **66** to provide massage therapy to the affected area. As the fluid is directed into the channels **66** at the proximal end **56**, the fluid travels along the channels **66** to the distal end **40** of the flexible member **14**, as indicated by arrow **72**, and adjust the distal end **40** of the channels **66** to the deployed state. As the distal end **40** is adjusted to the deployed state, the channels **66** will inflate from the distal end **40** to the proximal end **56**. The amount of fluid directed into the channels **66** may affect the amount of pressure applied to the affected area.

[0061] Based on the helix shape of the flexible member **14**, at the distal end **40** of the flexible member **14**, the channel **66A** may be disposed closer to the wrist than channel **66B**, and channel **66B** may be disposed closer to the wrist than channel **66C**. Accordingly, the channel **66A** may be adjusted to the deployed state first, followed by the channel **66B**, and then the channel **66C** to further contribute to a spiral pressure wave. This configuration produces the spiral pressure wave or a directional pattern of pressure from the wrist to the shoulder as indicated by arrow **74**. The spiral pressure wave pushes fluid up the arm toward the trunk of the body. The fluid is pushed out of the limb toward the heart to be combined with the circulatory system.

[0062] To adjust each channel **66** from the deployed state to the non-deployed state, the connection between the pump **68** and the channels **66** may be interrupted. Alternatively, the pump **68** may be configured to remove or vacuum the fluid from the channels **66**. The fluid may be stored by the treatment assembly **12** when the channels **66** are in the non-deployed state for use when subsequently adjusting the channels **66** to the deployed state.

[0063] Referring still to FIGS. 1-5, the treatment assembly **12** includes a control assembly **80** operably coupled to the proximal end **56** of the flexible member **14**. The control assembly **80** generally includes a controller **82**, a power source **84** (FIG. 18), and the pump **68**, all disposed within a housing **86**. The housing **86** may extend over the inlets **70** to position the pump **68** in fluid communication with the inlets **70** while obscuring any connection or tubing from view. Any connection or tubing may then be contained within the housing **86**, to provide a greater range of movement and flexibility to the patient of the flexible member **14**. The controller **82** is configured to activate and deactivate the pump **68** to adjust the channels **66** between the deployed state and the non-deployed state. The patient may connect the control assembly **80** to the flexible member **14** after applying the flexible member **14** to the affected area. Additionally or alternatively, the control assembly **80** may remain connected to the flexible member **14**.

[0064] The treatment assembly **12** includes a user-interface **90** for receiving inputs from the patient regarding the operation of the health management system **10**. For example, the user-interface **90**

may include buttons **92** on the housing **86**. The patient may input a command into the user-interface **90**, which may be communicated to the controller **82**. In response to the input, the controller **82** may activate or deactivate the pump **68**. A specific massage therapy protocol or an adjustment to the current massage therapy protocol may be controlled through the user-interface **90**.

[0065] Referring still to FIGS. **1-5**, conventional wraps may be time-consuming to wrap, unwrap, and rewrap around the affected area for each treatment. The therapeutic wraps **38, 50** disclosed herein may retain the helix shape through the engagement between the first coupling feature **18** and the second coupling feature **24** while sliding the therapeutic wraps **38, 50** on and off the affected area. The actuators **16** are configured to adjust from the deployed state to push fluid out of the affected area to recombine with the circulatory system.

[0066] With reference to FIGS. **6** and **7**, an additional or alternative configuration of the treatment assembly **12** is illustrated where the flexible member **14** is configured as a garment, such as a sock **100**. The sock **100** may be operably coupled with the actuators **16**, which are each configured as an electromagnetic linear actuator, such as a voice coil **102**. It is contemplated that the actuators **16** may be, for example, voice coils **102** with a moving magnet, voice coils **102** with a moving coil, speaker coils, any other electromagnetic linear actuator, or any other type of electromagnetic actuator. Each voice coil **102** is adjustable between the non-deployed state and the deployed state. The non-deployed state may be a retracted condition of each voice coil **102**, and the deployed state may be an extended condition of each voice coil **102** to apply pressure to the affected area. The voice coils **102** may be arranged in any practicable pattern or arrangement, such as in the illustrated voice coil group **102A-102D**, along the sock **100** to provide a directional pattern of pressure, as indicated by arrow **104**, from the foot toward the knee or hip of the patient. For example, the voice coils **102** may be arranged to substantially cover the sock **100** around the leg and top of the foot.

[0067] Each voice coil **102** is a linear actuator that extends and retracts to apply pressure to and remove pressure from the affected area. Generally, each voice coil **102** includes a coil assembly **106** and a magnetic field assembly **108**. The coil assembly **106** slides relative to the magnetic field assembly **108**, thereby adjusting the voice coil **102** between the deployed and non-deployed states. The current flowing through the coil assembly **106** may interact with a magnetic field generated by the magnetic field assembly **108** and generate a force vector. The force vector may be perpendicular to the direction of the current flowing through the coil assembly **106**, thereby moving the coil assembly **106** relative to the magnetic field assembly **108**. Reversing the movement of the coil assembly **106** may be accomplished through changing the polarity of the current flowing through the coil assembly **106**.

[0068] Each voice coil **102** may directly engage the skin of the patient. In such examples, the voice coils **102** are operably coupled to an inner surface of the sock **100**. Additionally or alternatively, the sock **100** may define apertures and the coil assembly **106** for each voice coil **102** may extend through a corresponding aperture when the voice coil **102** is in the deployed state to press against the skin of the patient. In another non-limiting example, the voice coils **102** may press against one layer of fabric of the sock **100** and press the layer of fabric against the skin of the patient (as best illustrated in FIG. **7**).

[0069] Referring still to FIGS. **6** and **7**, each voice coil **102** may be adjusted in response to a voltage supplied by the power source **84** (FIG. **18**). The voice coils **102** may be organized into voice coil groups, which are illustrated as the voice coil group **102A-102D**, and collectively referred to herein as the voice coils **102**. Depending on the length of the sock **100** or other garments, the sock **100** may include any practical number of groups of the voice coils **102**. The voice coils **102** may be configured to be activated sequentially from a distal portion of the leg to a proximal portion of the leg. Accordingly, the voice coil group **102A** arranged on the foot of the patient may be adjusted to the deployed state first. The voltage may be applied to the voice coil group **102A** to adjust the coil assemblies **106** to apply pressure to the foot, thereby driving fluid out of the foot toward the ankle.

[0070] The voice coil group **102B** disposed around the ankle may be adjusted to the deployed state next. The voice coil group **102B** may apply pressure to the skin of the ankle, thereby pressing fluid further up the leg away from the ankle. The voice coil group **102A** may remain in the deployed state to prevent fluid from returning to the foot in response to the pressure applied by the voice coil group **102B**. The voice coil group **102C** may be adjusted to the deployed state next to further drive fluid up the leg toward the knee. As the voice coil group **102C** is adjusted to the deployed state, the voice coil group **102A** may be adjusted to the non-deployed state to remove the pressure from the foot. The voice coil group **102B** may remain in the deployed state to prevent fluid from moving back toward the foot in response to the pressure applied by the voice coil group **102C**. The voice coil group **102D**, and any additional groups of voice coils **102**, may be configured to adjust to the deployed state in a similar sequential manner until reaching a proximal end of the sock **100**.

[0071] The sequential adjustment of the voice coils **102** to the deployed state generally provides a wave of pressure, as indicated by the arrow **104**. The direction of the pressure generally drives fluid out of the limb and to the central core of the body. The pressure wave may be continued over the period of time of the selected massage therapy protocol. The amount of pressure applied by each voice coil **102** may be proportional to the amount of voltage applied to the voice coil **102**, which may be adjusted through the user-interface **90** (FIG. **18**). Each coil group may be adjusted to the deployed state simultaneously or may be adjusted in a pattern from the distal voice coil **102** to the proximal voice coil **102**. While the treatment assembly **12** is illustrated as the sock **100**, it is contemplated that the treatment assembly **12** may be configured as other types of garments, including a sleeve or a shirt.

[0072] As illustrated in FIG. **7**, the voice coils **102** may indirectly apply pressure to the skin of the patient by affecting pressure on an engagement layer **116**. The engagement layer **116** may be disposed between the skin of the patient and each voice coil **102**. When the voice coils **102** are in the deployed state, the coil assembly **106** may press into the engagement layer **116**, consequently, pressing the engagement layer **116** against the skin of the patient to apply pressure to the affected area. As the coil assembly **106** retracts to the non-deployed state, the engagement layer **116** may return to an original condition, thereby removing pressure from the skin of the patient.

[0073] The engagement layer **116** may have a variety of configurations. For example, the engagement layer **116** may include a fabric or other materials that form the sock **100**. The engagement layer **116** may be an additional layer of fabric coupled to the sock **100**. Additionally or alternatively, the engagement layer **116** may include a polymeric or elastomeric material, such as rubber. In another example, the engagement layer **116** may be a layer of gel or gel beads. In an additional example, the engagement layer **116** may include a bladder filled with a fluid or gas. In configurations, the voice coil **102** may shift the fluid within the bladder to an area adjacent to the voice coil **102** thereby applying pressure around the voice coil **102** rather than or in addition to directly between the voice coil **102** and the skin of the patient. Further, the engagement layer **116** may include a textured pattern that abuts the skin to massage the skin as the voice coil **102** adjusts to the deployed state.

[0074] Referring to FIG. **8**, in an additional or alternative configuration of the voice coil **102**, a guide member **122** may be operably coupled to the coil assembly **106** of each voice coil **102**. The guide member **122** generally includes flexible projections **124** extending away from the magnetic field assembly **108**. The guide member **122** is adjusted with the movement of the coil assembly **106**.

[0075] Referring still to FIG. **8**, as well as to FIGS. **9A** and **9B**, the flexible projections **124** are configured to apply directional pressure, as indicated by arrow **126**. Accordingly, the flexible projections **124** are configured to provide greater pressure in the direction **126** and minimal or no pressure in the direction opposite the direction **126**. The flexible projections **124** may provide uniform or varying pressure based on the shape and arrangement of the flexible projections **124**. In the illustrated configuration, the flexible projections **124** are arranged generally parallel to one

another and have similar shapes. However, other configurations are contemplated based on the pressure to be applied against the skin of the patient.

[0076] As illustrated in FIG. 9A, the flexible projections **124** are curved in the same direction to push against the skin in the direction **126** as the voice coil **102** is adjusted to the deployed state. The flexible projections **124** are arranged so the direction of the pressure is toward the trunk or central core of the body. Accordingly, the voice coils **102** with the guide member **122** are configured to press against the skin in two directions (e.g., against the skin and in the direction **126**).

[0077] As the voice coil **102** is retracted to the non-deployed state, as illustrated in FIG. 9B, the flexible projections **124** may slide against and move away from the skin. Due to the curvature of the flexible projections **124**, as the voice coil **102** is adjusted to the non-deployed state, the flexible projections **124** may not apply significant pressure in a direction opposite the direction **126** as the voice coil **102** is adjusted to the non-deployed state. Accordingly, the flexible projections **124** may provide greater resistance and pressure when moving in one direction (e.g., the direction **126**) against the skin compared to the opposing direction. In this way, the voice coils **102** massage the skin in a single direction, which promotes the movement of fluid in the direction **126** toward the central core of the body. It is contemplated that the flexible projections **124** may directly contact the skin of the patient, or alternatively may contact the engagement layer **116** (FIG. 7).

[0078] With reference to FIG. 10, the sock **100** may include a slide assembly **134** operably coupled to each voice coil **102** for translating force from a first direction, as applied by the voice coil **102**, to a second direction. The slide assembly **134** may include an engagement feature **136** having a slide surface **138** and a slide **140** configured to slidably engage the slide surface **138**.

[0079] The slide assembly **134** may be operable between a retracted condition and an extended condition based on the state of the voice coil **102**. The engagement feature **136** includes a guide **152** and an arm **154** extending from the slide surface **138** into an interior **156** of the slide **140**. The slide **140** defines two apertures **158**, **160**, with the guide **152** extending through one aperture **158** and the arm **154** extending to the other aperture **160**. Additionally or alternatively, the slide **140** defines a protrusion **162** extending from an inner surface into the interior **156**. A biasing member **164** may be operably coupled to each of the arm **154** and the protrusion **162**. As illustrated, the biasing member **164** is arranged generally parallel to the slide surface **138** of the engagement feature **136**. However, other configurations of the slide assembly **134** are contemplated without departing from the teachings herein.

[0080] The interface between the engagement feature **136** and the slide **140** may be diagonal or angled, such that both the engagement feature **136** and the slide **140** are substantially wedge-shaped. The wedge-shaped engagement feature **136** and slide **140** may mate to form a substantially cuboid or prism slide assembly **134**. Moreover, the mating wedge-shapes of the engagement feature **136** and the slide **140** provide for the sliding movement of the slide **140** along the slide surface **138** between retracted and extended conditions.

[0081] Referring again to FIG. 10, as well as FIGS. 11A and 11B, each voice coil **102** may be operably coupled with the slide assembly **134**. The slide assembly **134** is configured to adjust between the retracted condition and the extended condition. When the voice coil **102** is in the deployed state, as illustrated in FIG. 11A, the slide assembly **134** is adjusted to the extended condition with the slide **140** offset from the engagement feature **136**. The coil assembly **106** of the voice coil **102** presses onto the engagement feature **136**, thereby shifting the engagement feature **136** in a first direction, as indicated by arrow **166**, toward the skin of the patient. Due to the angled interface of the slide assembly **134**, as the engagement feature **136** is shifted in the direction **166**, the slide **140** adjusts in a second direction, indicated by arrow **168**, by sliding along the slide surface **138** until the slide assembly **134** is in the extended condition.

[0082] As illustrated in FIG. 11A, the second direction **168** of the slide **140** is substantially perpendicular to the first direction **166** of the force applied by the voice coil **102**. Accordingly, the slide assembly **134** may translate the force of the voice coil **102** from the direction **166** to the

direction **168** to apply a downward and sliding force on the skin of the patient. The directional pressure in the direction **168** is generally directed to the central core of the patient to drive fluid toward the circulatory system to be processed. The force applied by the voice coil **102** may overcome the biasing force of the biasing member **164** to shift the slide **140** relative to the engagement feature **136**. It is contemplated that the slide assembly **134** may be used in conjunction with the engagement layer **116** (FIG. 7). In such configurations, the slide assembly **134** may affect pressure downward and in the direction **168** on the engagement layer **116**, which consequently affects the same or similar pressure on the skin of the patient.

[0083] As the voice coil **102** is adjusted to the non-deployed state, as illustrated in FIG. **11B**, the slide assembly **134** may adjust to the retracted condition where the slide **140** is generally aligned with the engagement feature **136**. The biasing force of the biasing member **164** retracts the slide **140** to its original position. As the voice coil **102** is not applying pressure to the slide assembly **134**, the slide **140** may shift across the skin of the patient, applying minimal or no pressure. Accordingly, the voice coil **102** may engage the slide assembly **134** to apply pressure in the direction **166** into the skin and in the direction **168** along the skin to press fluid along the limb toward the trunk of the body. As the voice coil **102** is a linear actuator and applies pressure in a single direction, the use of the guide member **122** (FIGS. **8-9B**) and/or the slide assembly **134** may be advantageous for applying a directional pressure on the skin of the patient.

[0084] Referring to FIGS. **10-12**, the apertures **158**, **160** defined by the slide **140** may be elongated to form a track for the guide **152** and the arm **154**. A portion of the guide **152** within the interior **156** of the slide **140** may have an increased width to maintain the connection between the engagement feature **136** and the slide **140**. As illustrated in FIG. **12**, when the slide assembly **134** is in the retracted condition, the guide **152** is positioned on one side (e.g., the left side) of the aperture **158**, and the arm **154** is disposed on the same side of the aperture **160**. As the slide **140** adjusts to the extended condition, the apertures **158**, **160** adjust relative to the guide **152** and the arm **154**, respectively. The slide **140** may move until each of the guide **152** and the arm **154** are disposed on the opposing side of the respective apertures **158**, **160** (e.g., the right side). Accordingly, the apertures **158**, **160** may define the fully retracted condition and the fully extended condition of the slide assembly **134**.

[0085] Referring to FIG. **13**, an additional or alternative configuration the health management system **10** includes the treatment assembly **12** configured as a support feature, such as a therapeutic pad **180**. The therapeutic pad **180** is generally a cushion or foam block that may be placed on a floor, a bed, or other furniture, or alternatively may be a mattress or mattress pad. The therapeutic pad **180** defines elongated cavities **182**, **184** configured to receive the legs (e.g., the affected area) of the patient. The therapeutic pad **180** may provide a massage therapy treatment to the patient who is resting on the therapeutic pad **180**. The control assembly **80** with the user-interface **90** may be disposed on the surface of the therapeutic pad **180** within the reach of the patient when the patient is resting on the therapeutic pad **180**. The therapeutic pad **180** may provide for a passive treatment system that provides a massage therapy protocol as the patient rests on the therapeutic pad **180** for a period of time. Other configurations of the therapeutic pad **180**, for example with a single elongated cavity **182** or for receiving an arm, are contemplated without departing from the teachings herein.

[0086] Referring to FIG. **14**, the treatment assembly **12** includes the actuators **16** configured as massage devices **186**. The massage devices **186** may be operably coupled with the therapeutic pad **180** within the cavities **182**, **184**. The massage devices **186** are operable between the deployed state and the non-deployed state. The deployed state may be an activated condition of each massage device **186**, and the non-deployed state may be a deactivated condition of each massage device **186**. When activated, each massage device **186** may apply pressure to the affected area.

[0087] The massage devices **186** may be placed in a selected position on the therapeutic pad **180** by the patient to provide a customizable treatment. The massage devices **186** may be arranged to apply

pressure to each side or the back of each leg placed in the cavities **182**, **184** depending on the location of the massage devices **186** as arranged by the user. The massage devices **186** may be coupled to the therapeutic pad **180** with snap features, Velcro®, hook-and-loop fasteners, or other fastening assemblies. Additionally or alternatively, the therapeutic pad **180** may define pockets to receive the massage devices **186**.

[0088] The therapeutic pad **180** may include coupling features extending over a substantial portion, or the entire, surface in the cavities **182**, **184** to allow the patient to couple the massage devices **186** at any location within the cavities **182**, **184** to provide a customizable treatment for the user. In this way, the surface of the cavities **182**, **184** may be substantially covered with coupling features to allow the massage devices **186** to be placed at any location along the cavities **182**, **184**. A covering may be used to conceal the massage devices **186** and the coupling features. It is also contemplated that the therapeutic pad **180** may include features on discrete portions of the surface within the cavities **182**, **184** that provide specific locations for coupling the massage devices **186** to the therapeutic pad **180**.

[0089] Referring to FIG. **15**, an exemplary configuration of the massage device **186** is illustrated. The massage device **186** is configured as a pressure assembly **188** that applies directional pressure to the affected area. The pressure assembly **188** includes a support structure **190** configured to couple to the therapeutic pad **180** and a plurality of elongated projections **192**, **194**, **196** operably coupled to the support structure **190**. The elongated projections **192**, **194**, **196** are disposed parallel to one another. Each elongated projection **192**, **194**, **196** is curved to apply greater pressure and resistance when rotating in one direction relative to the opposing direction.

[0090] In various examples, each elongated projection **192**, **194**, **196** is configured to rotate 360°. The elongated projection **192** is configured to rotate about a rotational axis a.sub.1 in the direction of arrow b.sub.1. Similarly, the elongated projection **194** is configured to rotate about a rotational axis a.sub.2 in the direction of arrow b.sub.2, and the elongated projection **196** is configured to rotate about a rotational axis a.sub.3 in the direction of arrow b.sub.3. In such configurations, the elongated projections **192**, **194**, **196** may apply the directional pressure when engaging the skin of the user. As each elongated projection **192**, **194**, **196** rotates, the elongated projections **192**, **194**, **196** will disengage the skin and rotate within the support structure **190** until once again engaging the skin. Additionally or alternatively, each elongated projection **192**, **194**, **196** may rotate about 180°. In such configurations, the elongated projections **192**, **194**, **196** apply a directional pressure when rotating in a first direction (e.g., in the direction of arrows b.sub.1, b.sub.2, b.sub.3, respectively) and apply minimal or no pressure when rotating in the opposing direction (e.g., in the direction opposite of arrows b.sub.1, b.sub.2, b.sub.3, respectively) due to the curvature of the elongated projections **192**, **194**, **196**. The elongated projections **192**, **194**, **196** may apply directional pressure to push fluid in a single direction, generally toward the trunk of the body to the circulatory system.

[0091] The pressure assembly **188** may include a motorized assembly **198** operably coupled to each elongated projection **192**, **194**, **196**. The motorized assembly **198** may rotate the elongated projections **192**, **194**, **196** simultaneously or in a pattern based on the direction the elongated projections **192**, **194**, **196** are applying pressure. The motorized assembly **198** may be advantageous for automatically rotating the elongated projections **192**, **194**, **196** when the massage device **186** is in the deployed state. The motorized assembly **198** may include, for example, a motor and gears for rotating each of the elongated projections **192**, **194**, **196** simultaneously.

[0092] Referring to FIG. **16**, an additional or alternative configuration of the massage device **186** is illustrated. The massage device **186** is configured as a pressure assembly **200** for applying general pressure to the affected area. The pressure assembly **200** includes a support structure **202** configured to couple to the therapeutic pad **180** and a massage feature **204** operably coupled to the support structure **202**. The massage feature **204** is generally configured to slide along the longitudinal extent of the support structure **202**, as illustrated by arrow c. Additionally or

alternatively, the massage feature **204** may rotate about a rotational axis **d** as illustrated by arrow **e**, which may be generally perpendicular to the longitudinal extent of the support structure **202**.

[0093] The massage feature **204** includes protrusions **206** configured to engage the skin of the user. The massage feature **204** may include any practicable number of protrusions **206** of any practicable shape or size. The position of the protrusions **206** on the skin may be adjusted through the rotation and sliding movement of the massage feature **204**. The pressure assembly **200** may apply general pressure to the skin of the patient to provide a massage treatment, but may not apply directional pressure similar to the pressure assembly **188** illustrated in FIG. 15. Accordingly, the pressure assembly **200** may apply general pressure to massage the selected area, but may not direct the fluid within the affected area in a single direction.

[0094] The pressure assembly **200** may include a motorized assembly **208** configured to slide and rotate the massage feature **204**. The motorized assembly **208** is operably coupled to the massage feature **204** and the support structure **202**. The motorized assembly **208** may rotate or translate the massage feature **204** to apply pressure and massage the skin. The motorized assembly **208** may be advantageous for adjusting the massage feature **204** when the massage device **186** is in the deployed state. The motorized assembly **208** may include, for example, a motor operably coupled with gears and/or rails to adjust the massage feature **204**.

[0095] Referring to FIGS. 14-16, the massage devices **186** may be configured to be activated simultaneously or independently from one another based on the massage therapy protocol. The controller **82** may send a signal to each massage device **186** to apply pressure to the affected area. Each massage device **186** may include communication circuitry to receive the signal from the controller **82**. Each massage device **186** may be a remote device separable from the therapeutic pad **180** and communicatively coupled to the controller **82**.

[0096] Referring to FIG. 17, in an additional or alternative configuration the health management system **10** includes the treatment assembly **12** configured as a support feature, such as a therapeutic pad **218**. The therapeutic pad **218** is generally a cushion or foam block that may be placed on a floor, a bed, or other furniture, or alternatively may be a mattress or mattress pad. The therapeutic pad **218** defines the cavities **182**, **184** configured to receive the legs of the user. The therapeutic pad **218** includes the actuators **16** configured as bladders **220**. The bladders **220** are arranged generally parallel to one another across the cavities **182**, **184**. Each bladder **220** is configured to engage the sides and the back of a leg of the user. Accordingly, each bladder **220** is generally U-shaped. It is also contemplated that multiple bladders **220** may be arranged to form each U-shape, such that the patient may apply massage treatment to one side of the leg or the back of the leg independently.

[0097] Each bladder **220** is adjustable between the deployed state and the non-deployed state. The deployed state may be an inflated condition of each bladder **220**, and the non-deployed state may be a deflated condition of each bladder **220**. The bladders **220** may be sequentially adjusted in accordance with the selected massage therapy protocol. For example, the bladders **220** may be sequentially adjusted to the deployed state from the ankle to the hip of each leg.

[0098] The bladders **220** may be arranged in bladder groups **220A-220D**, which are collectively referred to as the bladders **220**. The bladder group **220A** adjacent to the ankle may be adjusted to the deployed state first, pushing fluid away from the foot and toward the knee of the user. The bladder group **220B** may then be adjusted to the deployed state to further push the fluid toward the trunk of the body. The bladder group **220A** may remain in the deployed state to prevent fluid from returning to the ankle area in response to the pressure applied by the bladder group **220B**.

[0099] The bladder group **220C** may then be adjusted to the deployed state, further driving the fluid up the leg toward the hip. As the bladder group **220C** is adjusted to the deployed state, the bladder group **220A** may be adjusted to the non-deployed state, thereby removing pressure from the ankle area. The bladder group **220D** may then be adjusted to the deployed state to apply pressure on the upper portion of the leg. The bladders **220** may continue to be adjusted between the deployed state and the non-deployed state in a similar sequential manner. Each bladder **220** within the bladder

groups **220A-220D** may be adjusted simultaneously, or alternatively may be adjusted in a sequential pattern from a distal portion to a proximal portion. The bladder groups **220A-220D** for each leg (e.g., in the cavities **182, 184**) may be adjusted simultaneously or independently of one another based on the massage therapy protocol. It is contemplated that additional groups of bladders **220** may be included in the therapeutic pad **218**.

[0100] The therapeutic pad **218** includes the pump **68** coupled thereto and in fluid communication with each bladder **220**. The pump **68** is configured to direct fluid into each bladder **220** to adjust the bladders **220** to the deployed state. The amount of pressure applied by the bladders **220** may be adjusted by the amount of fluid directed into the bladders **220**. To adjust the bladders **220** to the non-deployed state, the fluid communication between the bladders **220** and the pump **68** may be disrupted, or alternatively, the pump **68** may be configured to remove or vacuum the fluid from the bladders **220**. The pump **68**, as well as any tubing or connections extending between the pump **68** and each bladder **220**, may be contained within the therapeutic pad **218**, which may provide more convenient treatment for the user. Specifically, the patient may utilize the therapeutic pad **218** for a longer period of time without having to accommodate external tubes.

[0101] Referring to FIGS. **13-17**, the therapeutic pads **180, 218** are generally more passive treatment options. The patient may rest his or her legs within the cavities **182, 184** for treatment. The therapeutic pads **180, 218** may be utilized for a longer duration of time, for example, over a period of hours. This may be advantageous for allowing the patient to use the therapeutic pads **180, 218** overnight or while sleeping, thereby providing treatment over a longer duration. The actuators **16** may be adjusted to the deployed state and remain in the deployed state for a period of time, for example, for a period of hours. Additionally or alternatively, the actuators **16** may be adjusted to the deployed state for a predetermined amount of time, for example, 15 minutes or 30 minutes, before being adjusted to the non-deployed state. In such configurations, the massage therapy protocol may provide continuous intervals of pressure for a period of hours. It is contemplated that the voice coils **102**, as discussed previously herein, may be included in the therapeutic pads **180, 218**. It is also contemplated that the therapeutic pads **180, 218** may include a single cavity for being used for treatment on an arm of the user.

[0102] Referring to FIGS. **18-21**, in an additional or alternative configuration of the health management system **10**, the treatment assembly **12** includes a base or therapeutic pad **230** operably coupled with a cover **232**. The therapeutic pad **230** is generally a cushion or foam block that may be placed on a floor, a bed, or other furniture, or alternatively may be a mattress or mattress pad. The cover **232** is generally constructed of a similar material as the therapeutic pad **230**. The therapeutic pad **230** defines a cavity **234** for receiving the affected area, such as a leg or an arm, of the patient. The cover **232** is operable between an opened position for inserting and removing the limb from the cavity **234**, as illustrated in FIGS. **18** and **19**, and a closed position for providing treatment to the patient, as illustrated in FIG. **20**. The cover **232** is generally arcuate, such that when the cover **232** is in the closed position, the therapeutic pad **230** and the cover **232** extend entirely around the limb and defines a rounded space for the limb of the patient.

[0103] The cover **232** is pivotally coupled to the therapeutic pad **230** and operable between the opened and closed positions. A first edge **236** of the cover **232** remains engaged with the therapeutic pad **230** proximate an abutment surface **238** as the cover **232** is moved between the opened and closed positions. As best illustrated in FIG. **19**, the abutment surface **238** is disposed adjacent to the cavity **234**. When the cover **232** is in the opened position, an outer surface of the cover **232** rests against the abutment surface **238** to retain the cover **232** in the opened position. The abutment surface **238** defines the fully opened position and provides a stopping position of the cover **232**. While illustrated as two components, the therapeutic pad **230** and the cover **232** may be a single component with a hinge, a living hinge, or another similar structure to allow the movement of the cover **232** between the opened and closed positions.

[0104] A second edge **240** of the cover **232** selectively engages a mating surface **242** of the

therapeutic pad **230**. When in the opened position, the second edge **240** of the cover **232** is free of a direct engagement with the mating surface **242** of the therapeutic pad **230**. When in the closed position, the second edge **240** rests on the mating surface **242**, enclosing the limb within the treatment assembly **12**. The treatment assembly **12** may include a connector extending between the cover **232** and the therapeutic pad **230** to retain or lock the cover **232** in the closed position while treatment is provided.

[0105] Referring still to FIGS. **18-21**, the treatment assembly **12** with the therapeutic pad **230** and the cover **232** includes the flexible member **14** configured as a sling **244**, which includes the actuators **16** configured as bladders **246**, illustrated in bladder groups **246A-246C**. The sling **244** extends between the cover **232** and the therapeutic pad **230** and is configured to support the limb of the user. The sling **244** generally extends from the second edge **240** of the cover **232**, along the underside of the cover **232**, through the cavity **234**, and is operably coupled to the mating surface **242** of the therapeutic pad **230**. In this way, the sling **244** encircles the limb when the cover **232** is in the closed position. It is contemplated that the treatment assembly **12** may include a single sling **244**, or a plurality of slings **244** arranged parallel to one another along the treatment assembly **12**.

[0106] Referring still to FIGS. **18, 20, and 21**, the sling **244** is adjustable between a relaxed condition, as illustrated in FIG. **20**, and a fitted condition, as illustrated in FIG. **21**. When in the relaxed condition, the sling **244** generally fits loosely around the limb of the patient and may be spaced apart from the skin on the patient. When the sling **244** is in the fitted condition, the sling **244** extends along the skin of the patient in a snug or form fit state.

[0107] A first end **248** of the sling **244** is fixedly coupled to the second edge **240** of the cover **232**. A second end **250** of the sling **244** is operably coupled with the therapeutic pad **230**. As best illustrated in FIGS. **20 and 21**, the second end **250** of the sling **244** extends into an interior of the therapeutic pad **230** adjacent to the cavity **234**. The second end **250** is operably coupled with an adjustment device, such as rollers **252**, which are operably coupled to or disposed within the therapeutic pad **230**. The rollers **252** are operably coupled with a motor **254**. When the motor **254** is activated, the rollers **252** are configured to adjust the sling **244** to the fitted condition by rotating in the direction of arrows **f, g**, respectively, and consequently pulling the second end **250** of the sling **244** in the direction of arrow **256**. Generally, the rollers **252** spin inwards, towards one another and to shorten the length of the sling **244** in the cavity **234**. The rollers **252** continue to adjust the sling **244** until a predetermined tension is reached. When the sling **244** reaches the predetermined tension, the motor **254** is deactivated, the rollers **252** cease rotating in the direction **f, g**, and the sling **244** fits snugly around the limb of the user. While illustrated as the rollers **252**, it is contemplated that the adjusting device may be any device or structure configured to adjust the sling **244**.

[0108] The rollers **252** may be used to adjust the cover **232** to the closed position, as well as pull the sling **244** to the fitted condition. As the rollers **252** adjust the sling **244** to the fitted condition, the sling **244** pulls on the cover **232** and automatically adjusts the cover **232** to the closed position. When the sling **244** is pulled to the predetermined tension, the predetermined tension may retain the cover **232** in the closed position. It is contemplated that the patient may manually adjust the cover **232** from the opened position to the closed position prior to activation of the motor **254**.

[0109] Referring still to FIGS. **18, 20, and 21**, once the sling **244** is in the fitted condition, the controller **82** is configured to begin the massage therapy protocol. The massage therapy protocol may be started automatically in response to the controller **82** detecting the pre-determined tension has been reached, or may be started in response to a user input. The sling **244** includes the bladders **246** extending parallel to one another and arranged along the sling **244**. The bladders **246** are configured to extend around a circumference of the limb when the cover **232** is in the closed position. It is contemplated that the sling **244** may include a plurality of bladders **246**, or alternatively, a single segmented bladder **246** where each segment is independently adjustable.

[0110] The bladders **246** are in fluid communication with the pump **68**, which is stored within the

therapeutic pad **230**. The pump **68** may be connected to the bladders **246** by a manifold, tubing, or other connectors. The bladders **246** are adjustable between the deployed state and the non-deployed state. The deployed state may be an inflated condition of each bladder **246**, and the non-deployed state may be a deflated condition of each bladder **246**. The pump **68** generally directs fluid into the bladders **246** to adjust the bladders **246** from the non-deployed state to the deployed state. To adjust the bladders **246** to the non-deployed state, the pump **68** may actively remove the fluid from the bladders **246** (e.g., vacuum), or the fluid connection between the pump **68** and the bladders **246** may be disrupted. The therapeutic pad **230** may include storage for housing excess fluid for when the bladders **246** are in the non-deployed state. Moreover, the pump **68** may be in fluid communication with one or more valves for directing the fluid to selected bladders **246**.

[0111] In the illustrated configuration, the bladders **246** are arranged in the bladder groups **246A-246C**, collectively referred to as the bladders **246**. The bladder group **246A** is disposed around a distal portion of the limb (e.g., a wrist or an ankle). The bladder group **246A** is generally adjusted to the deployed state first, applying pressure to the skin to push fluid away from the hand or foot, respectively, and up the limb. The bladder group **246B** is subsequently adjusted to the deployed state to push the fluid further toward the trunk of the body. The bladder group **246A** generally remains in the deployed state to prevent the fluid from moving away from the trunk of the body due to the pressure applied by the bladder group **246B**. The bladder group **246C** may then be adjusted to the deployed state. While the bladder group **246C** is adjusted to the deployed state, the bladder group **246A** may be adjusted to the non-deployed condition, removing pressure from the distal portion of the limb.

[0112] The bladders **246** may continue to be adjusted between the deployed state and the non-deployed state in a similar sequential manner. The sequential activation of the bladders **246** from the wrist to the shoulder or the ankle to the hip may drive fluid toward the central cavity of the body for processing by the circulatory system. It is contemplated that each bladder **246** in the bladder groups **246A-246C** may be adjusted simultaneously or may be adjusted sequentially in a distal to a proximal direction. It is contemplated that additional or fewer bladders **246** or bladder groups may be included in the sling **244**.

[0113] After the massage therapy treatment is completed, the rollers **252** loosen the sling **244** to return the sling **244** to the relaxed condition, allowing the patient to adjust the cover **232** to the opened position. The rollers **252** are configured to rotate in an opposing direction (e.g., opposite the directions f, g, respectively), which draws the second end **250** of the sling **244** in a direction opposite of the arrow **256** thereby loosening the sling **244**. It is contemplated that the therapeutic pad **230** may include a guide member for facilitating the adjustment of the sling **244** within the therapeutic pad **230**. The motor **254**, and consequently the rollers **252**, may be activated in response to a user input from the user-interface **90**.

[0114] Referring still to FIGS. **18-21**, the treatment assembly **12** with the therapeutic pad **230** and the cover **232** includes the pump **68**, the controller **82**, the power source **84**, the rollers **252**, and the motor **254**, as well as any valves or tubing in fluid communication with the pump **68** and the bladders **246**, all disposed in the interior of the therapeutic pad **230**. This configuration may be advantageous for more convenient use by the patient with the operational components contained within the treatment assembly **12**. Moreover, the therapeutic pad **230** and the cover **232** may provide a more passive treatment option for receiving massage treatment therapy. Accordingly, the patient may rest on the therapeutic pad **230** and allow the massage therapy treatment to be conducted for a longer period of time. The motor **254**, as well as the massage treatment protocol, may be activated, deactivated, or adjusted through the user-interface **90**.

[0115] The treatment assembly **12** may be configured to determine a circumference of the limb of the patient to provide metrics as to the status of the condition or the efficiency of the massage therapy protocol. The treatment assembly **12** includes a measuring system **260**. The measuring system **260** generally includes a sensor **262** and at least one marker **264**. The sensor **262** is

generally disposed on the therapeutic pad **230** proximate mating surface **242** where the sling **244** extends into the therapeutic pad **230** to obtain data from the markers **264**. The markers **264** are generally disposed along the sling **244**. The markers **264** are disposed at intervals in a substantially linear arrangement between the first end **248** of the sling **244** and the second end **250**. The markers **264** may be configured as barcodes, numbers, symbols, a ruler, etc. The sensor **262** is configured as an optical sensor configured to obtain data from the markers **264** within a field of view of the sensor **262**.

[0116] When the sling **244** reaches the predetermined tension, the sensor **262** may obtain the data from the markers **264** within the field of view and communicate the obtained data to the controller **82**. The controller **82** is configured to determine a circumference of the limb based on the data obtained by the sensor **262**. For example, a length of the sling **244** from the first end **248** to the second end **250** may be stored by the controller **82**, and the sensed information may be compared to the stored data to determine the circumference. Additionally or alternatively, the sensor **262** may scan the marker **264**, which may be indicative of a specific measurement. In an additional example, the sensor **262** may be configured to send the change in the length of the sling **244** within the therapeutic pad **230** between the relaxed and fitted conditions. It is contemplated that the sensor **262** may be any type of sensor configured to obtain measurement data.

[0117] Referring to FIG. 22, an additional or alternative configuration of the treatment assembly **12** is illustrated, which includes a therapeutic pad **268** and a cover **270** adjustable between opened and closed positions. The treatment assembly **12** includes the actuators **16** configured as bands **272** spaced apart from one another and arranged in a parallel configuration along the treatment assembly **12**. The bands **272** are configured to extend from the second edge **240** of the cover **270**, around the limb, through the cavity **234**, and operably couple to the mating surface **242** of the therapeutic pad **268**. The bands **272** are configured to support the limb resting in the cavity **234** and completely wrap around the circumference of the limb when the cover **270** is in the closed position.

[0118] Each band **272** is operably coupled to a roller **274** or another adjusting device within the therapeutic pad **268**. The motor **254** is operably coupled to each roller **274** to adjust the bands **272** from the relaxed condition, loosely fitting around the limb of the patient, to the fitted condition, snugly fitting around the limb. The pulling on the bands **272** by the rollers **274** may also operate to adjust the cover **270** from the opened position to the closed position. The rollers **274** are configured to pull each band **272** to a predetermined tension in the fitted condition. The bands **272** may be advantageous for providing a snug fit to different parts of the limb that may have a different circumference (e.g., an ankle compared to a thigh).

[0119] When the cover **270** is in the closed position and the bands **272** are in the fitted condition, the bands **272** are additionally adjusted between the deployed state and the non-deployed state. The deployed state may be a contracted condition of each band **272**, and the non-deployed state may be a relaxed condition of each band **272**. Each of the bands **272** may be independently adjusted to the deployed state to apply pressure to the skin of the patient.

[0120] In the illustrated configuration, band **272A** is disposed adjacent to the distal portion of the limb, such as the wrist or the ankle, and is generally adjusted to the deployed state first to apply pressure to the affected area. Band **272B** may be adjusted to the deployed state next to drive fluid further up the limb toward the trunk of the body. The band **272A** remains in the deployed state as the band **272B** is adjusted to the deployed state to prevent the fluid from moving back into the limb, away from the trunk of the body. Band **272C** is disposed more proximal than bands **272A**, **272B**, and is adjusted to the deployed state next. As the band **272C** is adjusted, the band **272A** may be adjusted to the non-deployed condition to remove pressure from a more distal area of the limb. However, the band **272B** generally remains in the deployed state during the adjustment of the band **272C** to prevent the fluid from moving away from the trunk of the body. Bands **272D**, **272E** may be similarly sequentially adjusted to the deployed state. The sequential activation of the bands **272A**-**272E** drives fluid toward the central cavity of the body for processing by the circulatory system.

The bands **272** may be a flexible material, such as cloth or fabric, or may be a metal or metal alloy material. Additional or fewer bands **272** may be included in the treatment assembly **12**.

[0121] Referring still to FIG. **22**, the treatment assembly **12** may include the measuring system **260**. The sensor **262** is disposed on the therapeutic pad **268** and the markers **264** are disposed on at least one of the bands **272**. The measuring system **260** is configured to obtain the data relating to the circumference of the limb, as described herein. It is also contemplated that the treatment assembly **12** may include markers **264** on each band **272** and a sensor **262** associated with each band **272**. In this way, different circumference measurements may be obtained at different portions of the limb.

[0122] Referring to FIGS. **23** and **24**, in an additional or alternative configuration of the health management system **10**, the treatment assembly **12** includes the flexible member **14** configured as a garment **276** having actuators **16** configured as bladders **278**, arranged in the illustrated bladder groups **278A-278E**. The garment **276** is illustrated as a sock that extends from the foot of the patient to the hip of the patient. The bladders **278** each extend around a circumference of the affected area (e.g., the leg).

[0123] The bladders **278** are operable between the deployed state, which may be an inflated condition, and the non-deployed state, which may be a deflated state. The bladders **278** are configured to drive fluid away from the foot and toward the trunk of the body. Accordingly, the bladder group **278A** disposed proximate to the foot is adjustable to the deployed state first. The bladder group **278B** may then be adjusted to the deployed state to further push the fluid toward the trunk of the body. The bladder group **278A** may remain in the deployed state to prevent fluid from returning to the ankle area in response to the pressure applied by the bladder group **278B**.

[0124] The bladder group **278C** may then be adjusted to the deployed state, further driving the fluid up the leg toward the hip. As the bladder group **278C** is adjusted to the deployed state, the bladder group **278A** may be adjusted to the non-deployed state, thereby removing pressure from the ankle area. The bladder groups **278D**, **278E** may be adjusted between the deployed state and the non-deployed state in a similar sequential manner as described herein. Each bladder **278** within the bladder groups **278A-278E** may be adjusted simultaneously, or alternatively may be adjusted in a sequential pattern from a distal portion to a proximal portion. The bladders **278** are sequentially adjusted to apply pressure to drive fluid from the limb.

[0125] The treatment assembly **12** also includes a therapeutic pad **280**. The therapeutic pad **280** is generally operable between a deflated state, as illustrated in FIG. **23**, and an inflated state, as illustrated in FIG. **24**. When in the inflated state, the therapeutic pad **280** assists the patient in lifting and elevating the limb. Convention devices often require the patient to place a pillow under the affected limb to elevate the limb and assist in lymphatic draining. This activity can be difficult for patients with higher weights or have larger fluid-filled limbs, causing the patient to have difficulty with mobility. The treatment assembly **12** with the therapeutic pad **280** assists the patient in lifting his or her limb for treatment.

[0126] The treatment assembly **12** illustrated in FIGS. **23** and **24** provides a two-stage inflation process. The treatment assembly **12** includes the pump **68** in fluid communication with the bladders **278** and the therapeutic pad **280**. When the patient applies the garment **276** to the affected area and activates the pump **68**, a first stage of inflation causes the bladders **278** to adjust to the deployed state to provide pressure to the limb. A second stage of inflation, after a predetermined amount of time or in response to a user input, causes the therapeutic pad **280** to be adjusted to the inflated condition. Generally, the therapeutic pad **280** is wedge-shaped, elevating the limb while providing comfort to the patient. When the limb is elevated, gravitational assistance is provided to aid in drainage of built-up lymphatic fluid inside the limb.

[0127] Generally, in the first stage of inflation, the bladders **278** may all be adjusted to the deployed state to provide an even or consistently distributed pressure on the limb. The even pressure may be maintained as the limb is elevated by the inflation of the therapeutic pad **280** in the

second stage of inflation. Once the therapeutic pad **280** is inflated, the bladders **278** may be adjusted to provide the sequential pressure to drive fluid in a proximal to distal direction to release the lymphatic fluid from the tissue. The therapeutic pad **280** allows the patient to elevate the affected limb for more efficient lymphatic draining without having to physically exert themselves. [0128] Referring still to FIGS. **23** and **24**, the garment **276** is generally coupled to the therapeutic pad **280** such that applying the garment **276** to the affected area positions the therapeutic pad **280** under the affected area. The pump **68** is in fluid communication with the bladders **278** and the therapeutic pad **280** via a manifold or tubing as well as valves to direct the fluid to a selected location. The pump **68** may direct fluid to the bladders **278** and the therapeutic pad **280**.

Additionally, the fluid communication between the bladders **278** and the therapeutic pad **280** may be disrupted to adjust each of the bladders **278** and the therapeutic pad **280** to the deflated state. Alternatively, the pump **68** may be configured to vacuum the fluid from the bladders **278** and/or the therapeutic pad **280**. The treatment assembly **12** is illustrated on the leg of the patient, but may also be used on the arm or other affected area without departing from the teachings herein.

[0129] Referring to FIG. **25**, the health management system **10** includes the controller **82** that has a processor **284**, a memory **286**, and other control circuitry. Instructions or routines **288** are stored within the memory **286** and executable by the processor **284**. The controller **82** includes one or more routines **288** for implementing the massage therapy protocols. The patient may select a specific massage therapy protocol through the user-interface **90**. The controller **82** may then adjust the actuators **16** between the deployed state and the non-deployed state in accordance with the selected massage therapy protocol.

[0130] The controller **82** disclosed herein may include various types of control circuitry, digital or analog, and may include the processor **284**, a microcontroller, an application specific circuit (ASIC), or other circuitry configured to perform the various input or output, control, analysis, or other functions described herein. The memory **286** described herein may be implemented in a variety of volatile and nonvolatile memory formats. The routines **288** include operating instructions to enable various methods and functions described herein.

[0131] For example, when the treatment assembly **12** includes the therapeutic wraps **38**, **50**, as illustrated in FIGS. **1-5**, the controller **82** may send a signal to the pump **68** to insert or remove air from the channels **66**. In examples where the treatment assembly **12** includes the sock **100**, as illustrated in FIGS. **6-12**, the controller **82** may be configured to send a current from the power source **84** to the voice coils **102** to adjust the voice coils **102** between the deployed state and the non-deployed state.

[0132] In examples where the treatment assembly **12** includes the therapeutic pad **180**, as illustrated in FIGS. **13-16**, the controller **82** may be configured to send a signal to the massage devices **186** through a wired or wireless communication mechanism, as will be discussed in further detail herein. The controller **82** may include communication circuitry **290** configured for bidirectional communication with the massage devices **186**. The controller **82** may send a signal to the respective motorized assembly **198**, **208** to adjust the massage devices **186** to the deployed state. Further, in examples where the treatment assembly **12** includes the therapeutic pad **218**, as illustrated in FIG. **17**, the controller **82** may be configured to activate the pump **68** to adjust the bladders **220** between the deployed state and the non-deployed state. Each configuration of the actuators **16** is configured to provide directional pressure to drive fluid toward the central core of the body.

[0133] In examples where the treatment assembly **12** includes the therapeutic pad **230**, as illustrated in FIGS. **18-22**, the controller **82** may be configured to send a signal to the pump **68** to add or remove fluid from the bladders **246**. In examples where the treatment assembly **12** includes the therapeutic pad **268**, as illustrated in FIG. **22**, the controller **82** may be configured to send a signal to the rollers **274** to adjust the bands **272** to the deployed state. In examples where the treatment assembly **12** includes the therapeutic pad **280**, as illustrated in FIGS. **23** and **24**, the

controller **82** may be configured to send a signal to the pump **68** to add or remove fluid from the bladders **278** and the therapeutic pad **280**.

[0134] The controller **82** includes the communication circuitry **290** configured to communicate with the massage devices **186**, as well as a remote device **294** included in the health management system **10**. The controller **82** may communicate with the massage devices **186**, the remote device **294**, and/or remote servers (e.g., cloud servers, Internet-connected databases, computers, etc.) via a communication interface **296**. The communication interface **296** may be a network having one or more various wired or wireless communication mechanisms, including any combination of wired (e.g., cable and fiber) or wireless communications and any network topology or topologies.

[0135] Exemplary communication networks include wireless communication networks, such as, for example, Bluetooth®, ZigBee®, Wi-Fi, IrDA, RFID, etc. The controller **82**, the massage devices **186**, and the remote device **294** may include circuitry configured for bidirectional wireless communication. Additional exemplary communication networks include local area networks (LAN) or wide area networks (WAN), including the Internet and other data communications services. It is contemplated that the controller **82**, the massage devices **186**, and the remote device **294** may communicate by any suitable technology for exchanging data.

[0136] The remote device **294** may be a remote handheld unit such as, for example, a phone, a tablet, a portable computer, a wearable device, etc. In a non-limiting example, the remote device **294** may be associated with a medical professional through a patient database system. Information relating to the massage therapy protocols may be communicated through the communication interface **296** to the patient database system. The medical professional may also assign massage therapy protocols or treatments through the communication interface **296**.

[0137] Additionally or alternatively, the remote device **294** may belong to the patient, thereby allowing the patient to control the treatment of his or her condition. The patient controls the selection, activation, and modification of the massage therapy protocol through the remote device **294**. Accordingly, the patient may control the actuators **16** through the remote device **294**. The patient may also control the amount of pressure applied by the actuators **16** to the affected area.

[0138] Referring still to FIG. **25**, the health management system **10** may include one or more health sensors **300** operably coupled to the treatment assembly **12**. The health sensors **300** may be configured to obtain one or more metrics from the patient of the treatment assembly **12**. The health sensors **300** may include a photoplethysmogram (PPG) sensor **302**. The PPG sensor **302** may be used to determine pulse oximetry to measure oxygen saturation levels or SpO₂ levels of the blood. Generally, the PPG sensor **302** may include optical sensors, which include a first light source configured to emit visible light (e.g., having a wavelength in a range between about 380 nm and about 700 nm), which can be white light (e.g., having a wavelength in a range between about 400 nm and about 700 nm) or red light (e.g., having a wavelength in a range between about 620 nm and about 750 nm) and a second light source configured to emit infrared light (e.g., having a wavelength in a range between about 700 nm and about 1050 nm). The two light sources may be advantageous as red light may be primarily absorbed by deoxygenated blood and infrared light may be primarily absorbed by oxygenated blood.

[0139] The PPG sensor **302** may also include a detector, such as a photodiode, configured to receive the light. The PPG sensor **302** may be utilized to monitor peaks, often called amplitudes, of the pulse. The metrics and data detected by the PPG sensor **302** may be communicated to the controller **82** to determine the percentage of oxygen in the blood.

[0140] Additionally or alternatively, the health sensors **300** may include a bioimpedance (BI) sensor **304**. Bioimpedance is a measure of how well the body impedes electrical current flow.

[0141] Impedance is measured through the application of a small electric current. The change in the measured voltage compared to the input voltage may determine the composition of the measured area. Bioimpedance spectroscopy may be used to measure the impedance of biological tissues at a series of frequencies, which may measure the fluid within cells and fluid outside of cells in the

measured area.

[0142] The BI sensor **304** may be configured as electrodes placed in contact with the skin of the patient, one of which emits a series of frequencies into the body. The frequencies penetrate certain aspects of the body, but not others. One of the electrodes may apply a small electric current from the power source **84** to be detected by the other electrode. Based on the penetration of the frequencies, the body composition of the patient may be determined. Using the data collected by the BI sensor **304**, a fluid level of the affected area may be obtained. Accordingly, utilizing the BI sensor **304**, the amount of fluid within the cells and outside of the cells in the affected area may be determined. The BI sensor **304** may be advantageous for determining an impedance of a specific limb, for example, the leg or the arm. The impedance measurement may be communicated to the controller **82**.

[0143] The fluid levels inside the cell compared to outside the cell may be advantageous for monitoring the condition of the patient. The metrics relating to the fluid levels of the affected area may also be monitored to determine the effectiveness of a massage therapy protocol. Sensing the amount of fluid using bioimpedance allows for the health management system **10** to determine the amount of time needed to spend using the therapy device for treatment. This creates an individualized treatment plan as each patient can range in severity in the condition and when or how often the condition flares up.

[0144] In another example, the health sensors **300** may include an electrocardiogram (ECG) sensor **306**. The ECG sensor **306** may be utilized to measure one or more of the six (6) lead ECG values depending on the configuration of the treatment assembly **12** or the health management system **10**. The ECG sensor **306** may include the electrodes contactable by the user. The ECG sensor **306** may provide a passive differential voltage measuring system. The data obtained by the ECG sensor **306** may be communicated to the controller **82**. It is contemplated that additional or alternative health sensors **300** that are configured to obtain one or more health metrics from the patient may be included in the health management system **10**.

[0145] Referring still to FIG. **25**, the data obtained by the health sensors **300** may be obtained for each affected area. The data may be utilized to determine various metrics, such as blood oxygen levels and fluid levels of the body. These metrics may be monitored to determine whether massage therapy is needed or the effectiveness of the massage therapy protocols. The health sensors **300** may be utilized to measure certain metrics for specific areas to monitor how each affected area, such as an arm or a leg, is responding to the massage therapy protocol.

[0146] The patient may monitor the metrics obtained by the health sensors **300** and the measuring system **260** through the remote device **294**. This may allow the patient to view and monitor the health metrics and adjust the massage therapy protocol in response to the metrics. The metrics obtained by the health sensors **300** and the measuring system **260** may allow the patient to monitor the condition being treated. Additionally or alternatively, the medical professional or caregiver associated with the patient database system may utilize the metrics to assign or vary the massage therapy protocol.

[0147] Referring to FIGS. **1-25**, the treatment assembly **12** provides directional pressure to the affected area to drive fluid toward the trunk of the body. The directional pressure may improve blood flow within the affected area. Additionally or alternatively, the directional pressure may drive the excess lymphatic fluid out of the limb and to the circulatory system to be processed. The directional pressure may result in directional activation of lymphatic vessels to assist in processing buildup of fluid in the body. The actuators **16** apply pressure from a distal portion of the body (e.g., a wrist or an ankle) to a proximal portion of the body (e.g., a shoulder or a hip), thereby activating the lymphatic vessels to transport fluid from the limb toward the trunk of the body. The directional pressure drives the fluid into the center core of the body to be processed by the circulatory system. The pressure may also assist in lymph node activation to assist in processing the lymphatic fluid. The health management system **10** may be advantageous for massaging lymph vessels and

activating lymph nodes to process the buildup of fluid in the affected area.

[0148] Use of the present device may provide for a variety of advantages. For example, the flexible member **14** may be configured to be worn by the patient to provide the massage therapy treatment. Additionally, the actuators **16** may apply pressure to the affected area to drive fluid within the affected area in a specific direction, pushing the fluid out of the limb toward the trunk of the body to be processed by the circulatory system. Further, the health management system **10** that includes the therapeutic pads **180, 218** may provide a passive system that may be used for longer periods of time. The therapeutic pads **180, 218** may provide treatment to a patient overnight or over a period of hours. The therapeutic pads **230, 268, 280** may provide additional passive treatment options.

Moreover, the measuring system **260** provides a process for obtaining data regarding the circumference of the limb within the treatment assembly **12**. Also, the patient may use the remote device **294** to control the actuators **16**, as well as to select or adjust the massage therapy protocol.

[0149] Additionally, the health management system **10** may include the health sensors **300** to obtain one or more health metrics from the patient. The patient may utilize the metrics to monitor the condition being treated, as well as the effectiveness of the treatment through the massage therapy protocol. The patient may also utilize the metrics obtained by the health sensors **300** to select a massage therapy protocol or a duration of the massage therapy treatment. Additional benefits or advantages may be realized and/or achieved.

[0150] The device disclosed herein is further summarized in the following paragraphs and is further characterized by combinations of any and all of the various aspects described therein.

[0151] According to an aspect of the present disclosure, a wrap for covering an affected area includes a flexible member configured to wrap around an affected area. Actuators are operably coupled to the flexible member. Each actuator is operable between a deployed state and a non-deployed state. A first coupling feature is disposed proximate a first edge on a first surface of the flexible member. A second coupling feature is disposed proximate a second edge on a second surface of the flexible member. The first coupling feature is configured to engage the second coupling feature when the flexible member is wrapped around the affected area and when the flexible member is slidably adjusted relative to the affected area. A retaining feature is coupled to at least one end of the flexible member and is configured to retain the flexible member in a selected position on the affected area.

[0152] According to another aspect, actuators are channels defined by a flexible member that extend along a longitudinal extent of the flexible member.

[0153] According to another aspect, a pump is in fluid communication with each channel via inlets defined by a flexible member. The pump is configured to direct fluid into each channel to provide pressure to an affected area.

[0154] According to another aspect, a flexible member forms a helix shape when a first coupling feature is engaged with a second coupling feature. The flexible member maintains the helix shape when the flexible member is slidably removed from the affected area and when the flexible member is slidably applied to an affected area.

[0155] According to another aspect, a retaining feature is an aperture defined in a distal end of a flexible member and is configured to receive a finger of a patient.

[0156] According to another aspect, a retaining feature includes a distal elongated flap having a first distal coupling feature that engages a second distal coupling feature around said affected area and a proximal elongated flap having a first proximal coupling feature that engages a second proximal coupling feature around said affected area.

[0157] According to another aspect of the present disclosure, a garment includes a flexible member configured to be worn on an affected area. A power source is operably coupled to the flexible member. An electromagnetic actuator is operably coupled to the power source. The electromagnetic actuator is operable between a deployed state and a non-deployed state in response to a voltage from the power source. The electromagnetic actuator applies pressure to the affected area when in

the deployed state.

[0158] According to another aspect, an electromagnetic actuator is a voice coil.

[0159] According to another aspect, an engagement layer is operably coupled to an electromagnetic actuator. The electromagnetic actuator presses against the engagement layer when in a deployed state and, consequently, applies pressure to an affected area via the engagement layer.

[0160] According to another aspect, a slide assembly is coupled to the electromagnetic actuator. The slide assembly is operable between an extended condition when the electromagnetic actuator is in the deployed state and a retracted condition when the electromagnetic actuator is in the non-deployed state. An electromagnetic actuator is configured to apply force to the slide assembly when in a deployed state and, consequently, adjust the slide assembly to the extended condition to apply directional pressure to an affected area.

[0161] According to another aspect, a guide member is operably coupled to an electromagnetic actuator and includes flexible projections. The flexible projections are configured to apply directional pressure to an affected area when the electromagnetic actuator is in the deployed state.

[0162] According to another aspect of the present disclosure, a health management system includes a therapeutic pad for supporting an affected area of a patient. Actuators are operably coupled to the therapeutic pad. A controller is communicatively coupled to the actuators. The controller is configured to adjust the actuators between a deployed state and a non-deployed state. The actuators are configured to apply pressure to the affected area supported by the therapeutic pad when the actuators are in the deployed state.

[0163] According to another aspect, a sensor is operably coupled to a therapeutic pad to obtain a health metric from a user. The sensor is at least one of a photoplethysmogram sensor, an electrocardiogram sensor, and a bioimpedance sensor.

[0164] According to another aspect, a controller is configured to adjust actuators between a deployed state and a non-deployed state in response to a massage therapy protocol. The massage therapy protocol extends over a period of hours.

[0165] According to another aspect, the actuators are bladders in fluid communication with a pump to be adjusted between a deployed state and a non-deployed state. The bladders are disposed in a therapeutic pad.

[0166] According to another aspect, actuators are remote massage devices selectively coupled to a surface of a therapeutic pad. The remote massage devices include at least one of elongated protrusions configured to rotate relative to a support structure and massage features configured to translate and rotate relative to the support structure.

[0167] According to another aspect, a cover is operably coupled to a therapeutic pad. The cover is adjustable between an opened position and a closed position relative to an elongated cavity defined by the therapeutic pad. A sling extends between the cover and the therapeutic pad, and the sling includes the actuators. Rollers are operably coupled to the therapeutic pad and the sling, and the rollers are configured to adjust the sling between a relaxed condition and a fitted condition.

[0168] According to another aspect, a flexible member is configured to be worn over an affected area. The flexible member is coupled to a therapeutic pad. Actuators are bladders disposed within the flexible member.

[0169] According to another aspect, a pump is in fluid communication with bladders and a therapeutic pad. The therapeutic pad is operable between a deflated state and an inflated state to selectively elevate the affected area.

[0170] According to another aspect, a controller is configured to adjust bladders to a deployed state to apply pressure to an affected area, adjust a therapeutic pad to an inflated state to elevate the affected area, and selectively and sequentially adjust the bladders between the deployed state and a non-deployed state to provide directional pressure along the affected area.

[0171] A means for treating that includes a means for wearing on an affected area. Means for applying pressure are operably coupled to the means for wearing. The means for applying pressure

are operable between a deployed state and a non-deployed state. A first means for coupling is disposed proximate a first edge on a first surface of the means for wearing. A second means for coupling is disposed proximate a second edge on a second surface of the means for wearing. The first means for coupling is configured to engage the second means for coupling when the means for wearing is on the affected area and when the means for wearing is slidably adjusted relative to the affected area. A means for retaining is coupled to at least one end of the means for wearing and configured to retain the means for wearing in a selected position on the affected area.

[0172] Related applications, for example those listed herein, are fully incorporated by reference. Descriptions within the related applications are intended to contribute to the description of the information disclosed herein as may be relied upon by a person of ordinary skill in the art. Any changes between any of the related applications and the present disclosure are not intended to limit the description of the information disclosed herein, including the claims. Accordingly, the present application includes the description of the information disclosed herein as well as the description of the information in any or all of the related applications.

[0173] It will be understood by one having ordinary skill in the art that construction of the described disclosure and other components is not limited to any specific material. Other exemplary embodiments of the disclosure disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

[0174] For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

[0175] It is also important to note that the construction and arrangement of the elements of the disclosure, as shown in the exemplary embodiments, are illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes, and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts, or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

[0176] It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

Claims

1. A therapeutic wrap for covering an affected area, comprising: a flexible member configured to wrap around said affected area; actuators operably coupled to the flexible member, wherein each actuator is operable between a deployed state and a non-deployed state; a first coupling feature disposed proximate a first edge on a first surface of the flexible member; a second coupling feature disposed proximate a second edge on a second surface of the flexible member, wherein the first coupling feature is configured to engage the second coupling feature when the flexible member is wrapped around said affected area and when the flexible member is slidably adjusted relative to said affected area; and a retaining feature coupled to at least one end of the flexible member and configured to retain the flexible member in a selected position on said affected area.
2. The therapeutic wrap of claim 1, wherein the actuators are channels defined by the flexible member that extend along a longitudinal extent of the flexible member.
3. The therapeutic wrap of claim 2, further comprising: a pump in fluid communication with each channel via inlets defined by the flexible member, wherein the pump is configured to direct fluid into each channel to provide pressure to said affected area.
4. The therapeutic wrap of claim 1, wherein the flexible member forms a helix shape when the first coupling feature is engaged with the second coupling feature, and wherein the flexible member maintains the helix shape when the flexible member is slidably removed from said affected area and when the flexible member is slidably applied to said affected area.
5. The therapeutic wrap of claim 1, wherein the retaining feature is an aperture defined in a distal end of the flexible member and configured to receive a finger of a patient.
6. The therapeutic wrap of claim 1, wherein the retaining feature includes a distal elongated flap having a first distal coupling feature that engages a second distal coupling feature around said affected area and a proximal elongated flap having a first proximal coupling feature that engages a second proximal coupling feature around said affected area.
7. A garment, comprising: a flexible member configured to be worn on an affected area; a power source operably coupled to the flexible member; and an electromagnetic actuator operably coupled to the power source, wherein the electromagnetic actuator is operable between a deployed state and a non-deployed state in response to a voltage from the power source, and wherein the electromagnetic actuator applies pressure to the affected area when in the deployed state.
8. The garment of claim 7, wherein the electromagnetic actuator is a voice coil.
9. The garment of claim 7, further comprising: an engagement layer operably coupled to the electromagnetic actuator, wherein the electromagnetic actuator presses against the engagement layer when in the deployed state and, consequently, applies the pressure to the affected area via the engagement layer.
10. The garment of claim 9, wherein the engagement layer is at least one of a fabric layer, a polymeric layer, an elastomeric layer, a layer of gel, a layer of gel beads, and a bladder.
11. The garment of claim 7, further comprising: a slide assembly coupled to the electromagnetic actuator, wherein the slide assembly is operable between an extended condition when the electromagnetic actuator is in the deployed state and a retracted condition when the electromagnetic actuator is in the non-deployed state, wherein the electromagnetic actuator is configured to apply force to the slide assembly when in the deployed state and, consequently, adjust the slide assembly to the extended condition to apply the pressure as directional pressure to the affected area.
12. The garment of claim 11, wherein the slide assembly includes: an engagement feature including a guide and an arm extending from a slide surface; a slide defining a protrusion extending into an interior thereof, wherein the guide and the arm extend into the interior of the slide; and a biasing member operably coupled with the arm and the protrusion to bias the slide assembly to the retracted condition.
13. The garment of claim 12, wherein the electromagnetic actuator is configured to apply the force

to the slide assembly in a first direction in the deployed state which causes movement of the slide in a second direction to the extended condition, the second direction being perpendicular to the first direction.

14. The garment of claim 7, further comprising: a guide member operably coupled to the electromagnetic actuator and including flexible projections, wherein the flexible projections are configured to apply the pressure as directional pressure to the affected area when the electromagnetic actuator is in the deployed state.

15. The garment of claim 14, wherein the flexible projections are curved in a single direction and extend parallel with one another to apply the directional pressure.

16. The garment of claim 7, wherein the non-deployed state is a retracted condition and the deployed state is an extended condition.

17. The garment of claim 7, wherein the electromagnetic actuator includes a plurality of electromagnetic linear actuators, and wherein the plurality of electromagnetic linear actuators are configured to be adjusted in sequence to provide the pressure as directional pressure to the affected area.

18. The garment of claim 7, wherein the electromagnetic actuator includes voice coils, and wherein each voice coil includes: a coil assembly; and a magnetic field assembly, wherein a current is configured to be directed through the coil assembly to interact with a magnetic field generated by the magnetic field assembly to move the coil assembly relative to the magnetic field assembly, thereby adjusting the voice coil between the deployed and non-deployed states.

19. The garment of claim 7, further comprising: a sensor operably coupled to the flexible member to obtain a health metric from a user, wherein the sensor is at least one of a photoplethysmogram sensor, an electrocardiogram sensor, and a bioimpedance sensor.

20. A health management system, comprising: a therapeutic pad having an upper surface for supporting an affected area of a patient thereon, wherein the therapeutic pad is operable between a deflated state and an inflated state where a distal end farther from a trunk of the patient is at an increased height relative to a proximal end closer to the trunk of the patient to selectively elevate the affected area; a flexible member coupled with the upper surface and configured to be worn over the affected area; bladders included in the flexible member and operable between a deployed inflated state and a non-deployed deflated state; a pump in fluid communication with the bladders and the therapeutic pad; and a controller communicatively coupled to the pump, wherein the controller is configured to adjust the bladders to the deployed inflated state in a first stage of inflation to apply pressure to the affected area supported by the therapeutic pad and adjust the therapeutic pad to the inflated state in a second stage of inflation to elevate the affected area.
